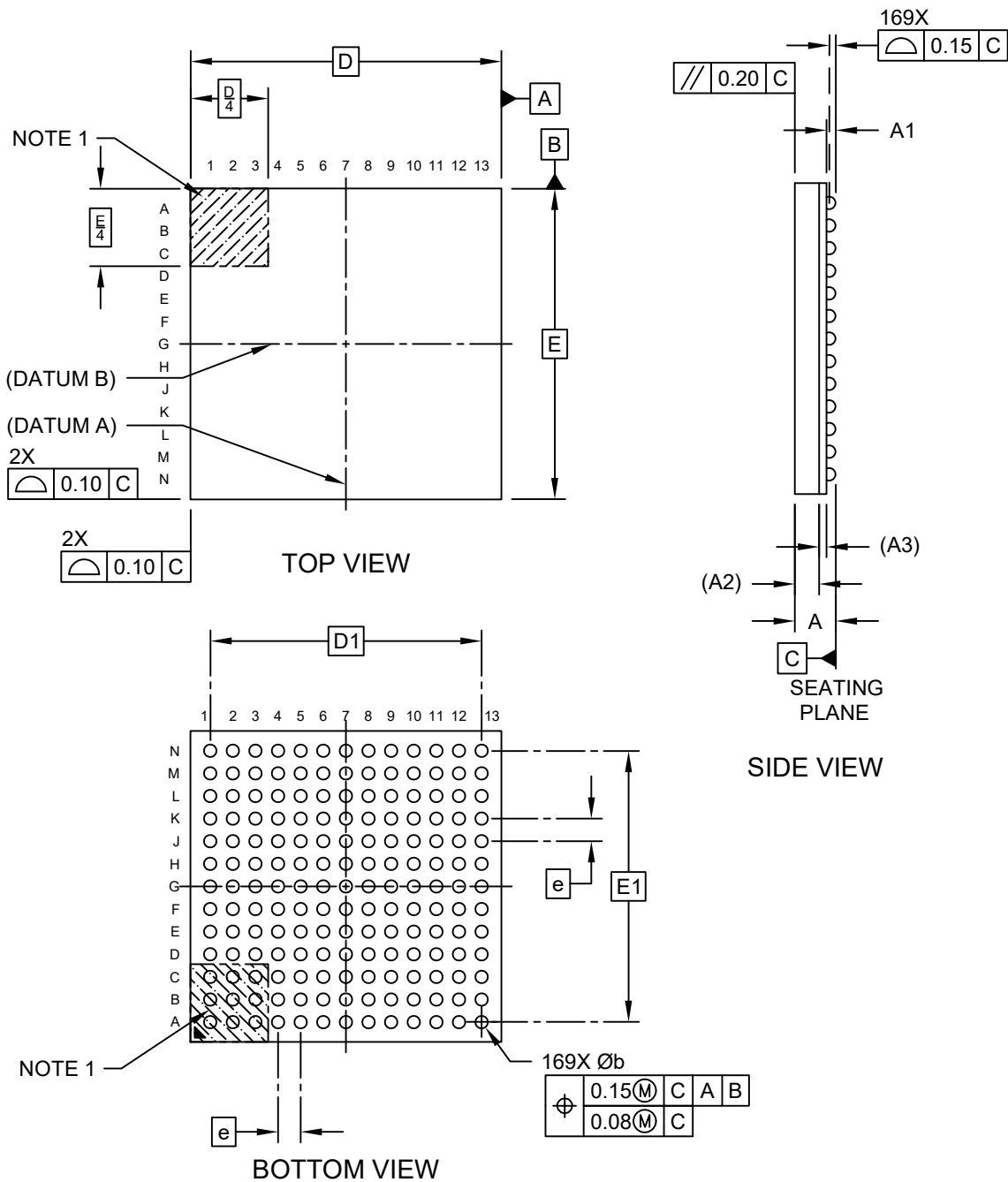


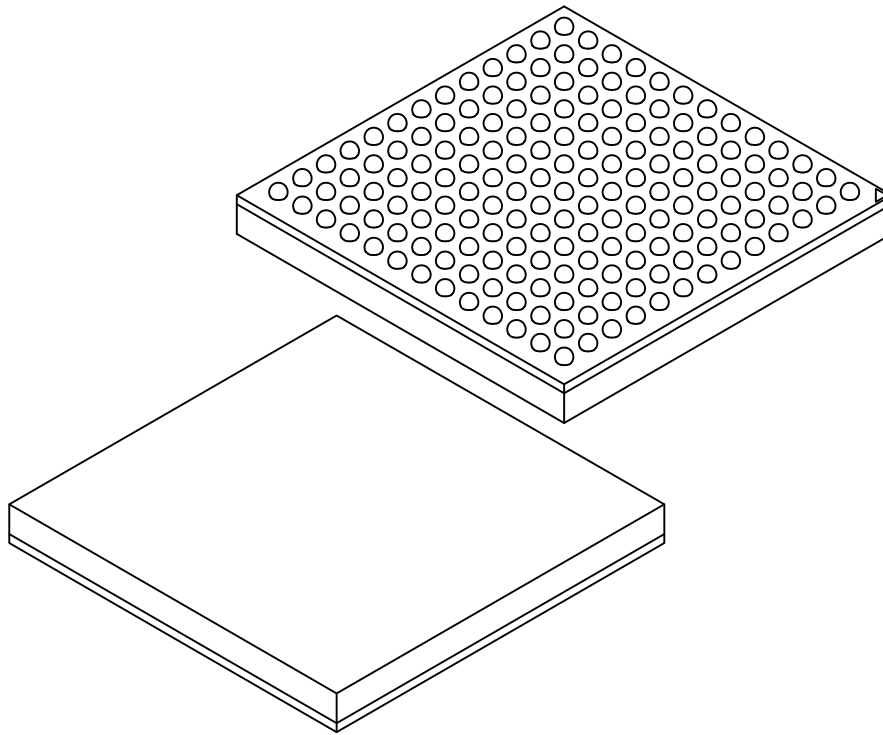
## 169-Ball Low Profile Ball Grid Array (6JX) - 11x11 mm Body [LFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## 169-Ball Low Profile Ball Grid Array (6JX) - 11x11 mm Body [LFBGA]

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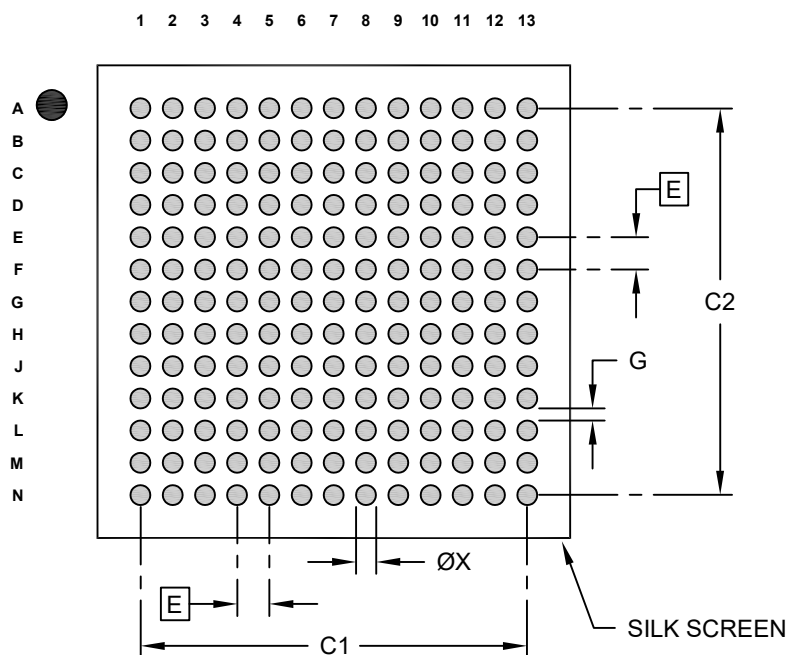
| Units                    |    | MILLIMETERS |       |      |
|--------------------------|----|-------------|-------|------|
| Dimension Limits         |    | MIN         | NOM   | MAX  |
| Number of Terminals      | N  | 169         |       |      |
| Pitch                    | e  | 0.80 BSC    |       |      |
| Overall Height           | A  | 1.33        | 1.445 | 1.56 |
| Standoff                 | A1 | 0.25        | -     | 0.40 |
| Mold Thickness           | A2 | 0.86 REF    |       |      |
| Substrate Thickness      | A3 | 0.26 REF    |       |      |
| Overall Length           | D  | 11.00 BSC   |       |      |
| Overall Terminal Spacing | D1 | 9.60 BSC    |       |      |
| Overall Width            | E  | 11.00 BSC   |       |      |
| Overall Terminal Spacing | E1 | 9.60 BSC    |       |      |
| Terminal Diameter        | b  | 0.40        | 0.45  | 0.50 |

**Notes:**

- Pin A1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

## 169-Ball Low Profile Ball Grid Array (6JX) - 11x11 mm Body [LFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



### RECOMMENDED LAND PATTERN

| Units                       |    | MILLIMETERS |      |      |
|-----------------------------|----|-------------|------|------|
| Dimension Limits            |    | MIN         | NOM  | MAX  |
| Contact Pitch               | E  | 0.80 BSC    |      |      |
| Overall Contact Pad Spacing | C1 |             | 9.60 |      |
| Overall Contact Pad Spacing | C2 |             | 9.60 |      |
| Contact Pad Width (X169)    | X1 |             |      | 0.50 |
| Contact Pad to Contact Pad  | G  | 0.30        |      |      |

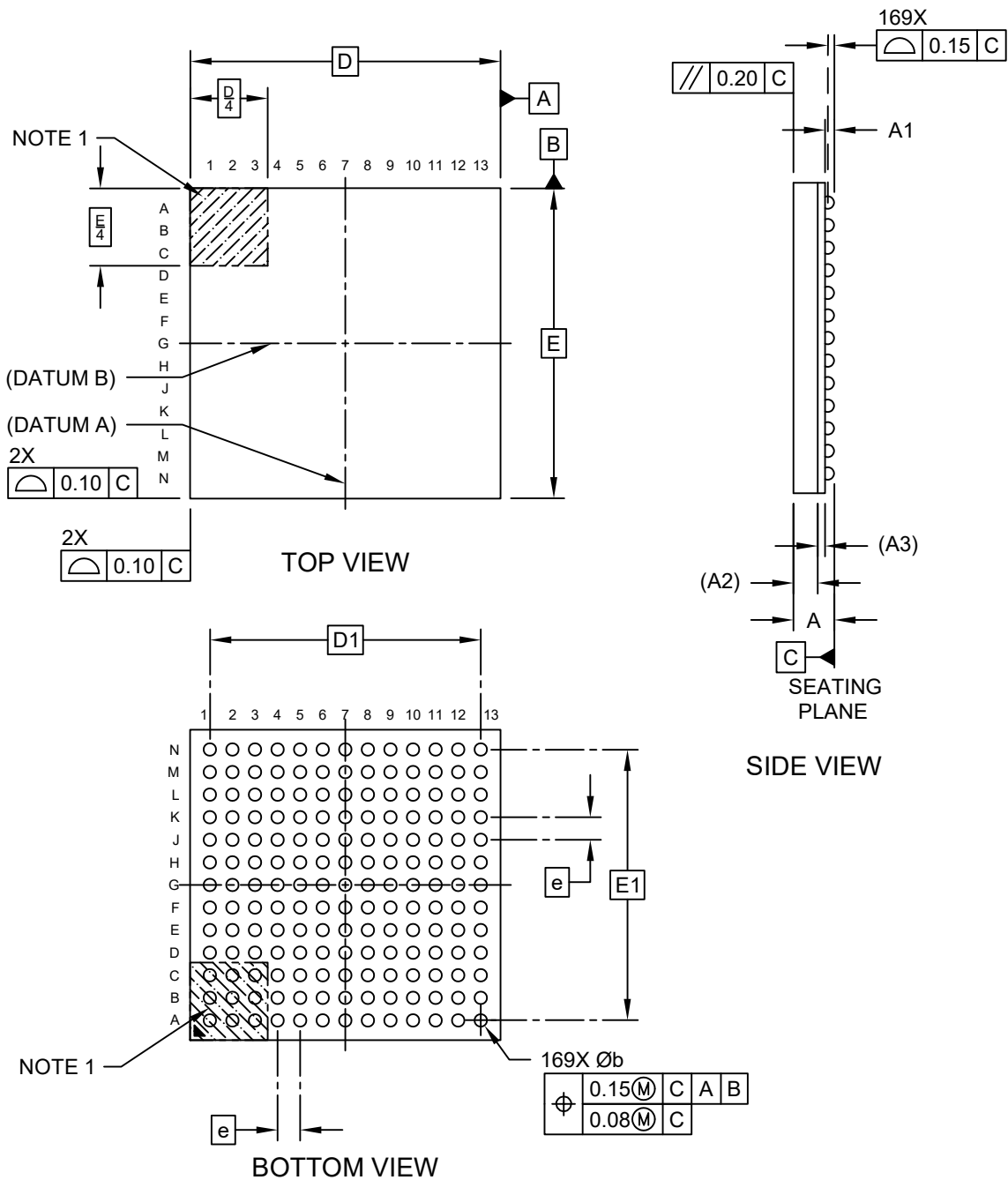
#### Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

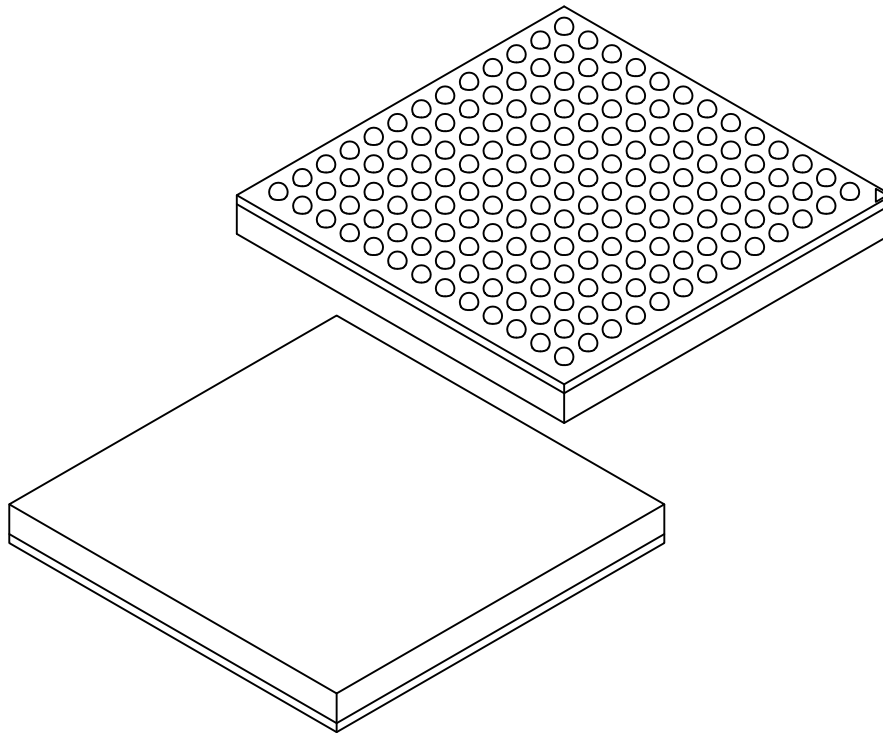
## 169-Ball Low Profile Ball Grid Array (6J) - 11x11 mm Body [LFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## 169-Ball Low Profile Ball Grid Array (6J) - 11x11 mm Body [LFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



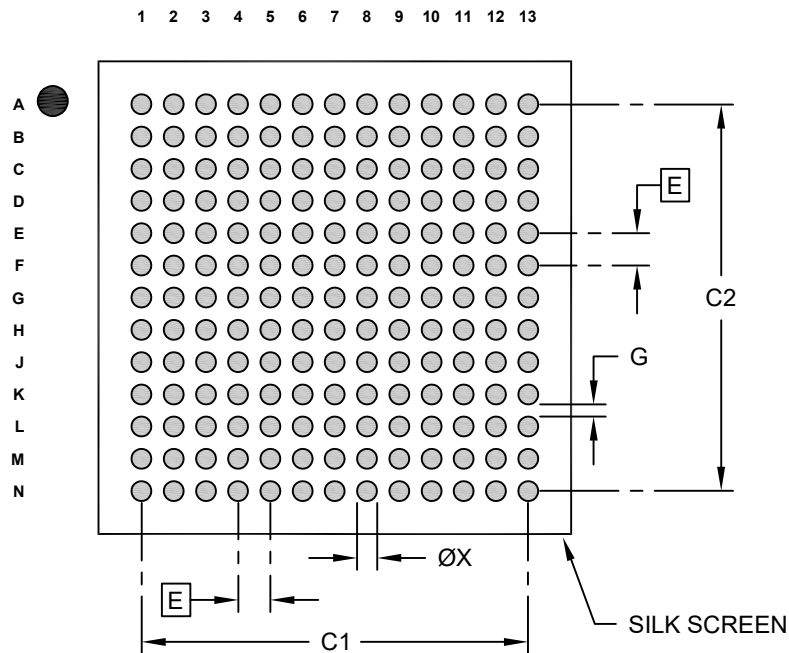
| Units                    |    | MILLIMETERS |       |      |
|--------------------------|----|-------------|-------|------|
| Dimension Limits         |    | MIN         | NOM   | MAX  |
| Number of Terminals      | N  | 169         |       |      |
| Pitch                    | e  | 0.80 BSC    |       |      |
| Overall Height           | A  | 1.33        | 1.445 | 1.56 |
| Standoff                 | A1 | 0.25        | -     | 0.40 |
| Mold Thickness           | A2 | 0.86 REF    |       |      |
| Substrate Thickness      | A3 | 0.26 REF    |       |      |
| Overall Length           | D  | 11.00 BSC   |       |      |
| Overall Terminal Spacing | D1 | 9.60 BSC    |       |      |
| Overall Width            | E  | 11.00 BSC   |       |      |
| Overall Terminal Spacing | E1 | 9.60 BSC    |       |      |
| Terminal Diameter        | b  | 0.40        | 0.45  | 0.50 |

### Notes:

1. Pin A1 visual index feature may vary, but must be located within the hatched area.
2. Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

## 169-Ball Low Profile Ball Grid Array (6J) - 11x11 mm Body [LFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

| Units                       |    | MILLIMETERS |      |      |
|-----------------------------|----|-------------|------|------|
| Dimension Limits            |    | MIN         | NOM  | MAX  |
| Contact Pitch               | E  | 0.80 BSC    |      |      |
| Overall Contact Pad Spacing | C1 |             | 9.60 |      |
| Overall Contact Pad Spacing | C2 |             | 9.60 |      |
| Contact Pad Width (X169)    | X1 |             |      | 0.50 |
| Contact Pad to Contact Pad  | G  | 0.30        |      |      |

**Notes:**

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.